💧 DropWise

More than two billion people still have no safely managed drinking water, and the problem feels too big and too far away to touch. DropWise makes it concrete: it shows where Earth's water really is, and lets you try the five fixes that work — in a field, a treatment plant, a desalination plant, a leaking pipe network and your own kitchen.

Built for **Sustainable Development Goal 6, Clean Water and Sanitation**. Entry for **Infomatrix 2027**, category **Software Development** (Project Contest), by **Nicole Puzur**, supervised by **Azamat Jayush**.

🔴 **Live app → *add the Vercel address here***

📦 **Source code → github.com/nicolepuzur/DropWise**

Infomatrix requirements checklist

Checked against the official regulations, the Software Development page and the evaluation rubric, read on 27 October 2026. The regulations state that rules can change without notice, so they are to be re-read before submitting.

Requirement	Where it is met
Entry and documentation in English (Regulations §5)	The site opens in English by default (Romanian and Russian are optional); all documentation is in English
At most two students and a supervisor, as distinct roles (§2, §5)	People involved: one student and one supervisor
Project description with idea sketches or images (§5)	Description, screenshots and the design plan
Early draft plans and designs (§5, §6)	Early-stage plan & design decisions and the design plan
Development iterations and a timeline (§5, §6)	Development timeline and the Git history
Lists of software and hardware (§5, §6)	Specifications
Names of contributors (§6)	People involved
Problems encountered and their solutions (§6)	Problems faced
Open source, not subject to anyone's proprietary rights (§10)	Public repository under the MIT licence
No third-party rights infringed (§13)	Third-party content & licences; no photographs or illustrations from other sources
Cite all prompts, models and AI resources, in the documentation and the presentation (Software Development page)	Responsible AI, the full AI usage log, and the site's *About* page
Documentation uploaded with the entry (§6)	This README with the AI usage log as an appendix, also as a PDF: docs/DropWise-documentation.pdf
A real-world problem, practical and user-friendly	The problem; the app works on desktop and phone

How the nine judging criteria are addressed

The jury scores nine criteria from 1 to 5. This is where the project answers each one; the scores are the jury's to give.

Criterion	What the project does
Originality / creativity	Each of the five solutions in the brief becomes a small model you can change; the planet is poured into 100 litres, with lakes and rivers drawn at their true size; the leak hunter is a game; the six hero scenes are drawn in code
Practicality	A real problem (SDG 6) with cited numbers; every tool ends in litres a person, farm or city could save; no account, no install, works on a phone
UI / UX	A full-screen slideshow, navigation on the right, three languages, a skip link, visible focus, keyboard control of the slideshow, and no forced motion for people who ask for less
Functionality	52 automated tests, inputs clamped and saved data re-validated, every page checked automatically in English, Romanian and Russian at desktop and phone size (layout and console errors) and by eye on the main pages; limits of the models are written next to each tool
Technical skills	React 19, TypeScript and Tailwind v4; pure, tested calculation modules; a custom i18n layer with a test that keeps three languages in step; SVG scenes from seeded random numbers; a pinned horizontal scroll row; validated local storage
Oral presentation	A 15-minute outline with the demo flow and likely questions is in docs/SUBMISSION.md
Documentation	This README and its PDF, the design plan, the AI usage log, screenshots, sources and a licence table
The stand design	A poster and booth plan is in docs/SUBMISSION.md
Responsible AI usage	Tool, model and every prompt cited; the checks that were run; a step-by-step process to reproduce the build; no AI-generated images; licensing and privacy covered (AI usage log)

Official dates (from infomatrix.ro): registration for Season 2027 opens on 1 January 2027, the submission deadline is **15 March 2027** (documentation, source code and a demo video), finalists are announced on **20 March 2027**, and the Global Finals are in Bucharest on **27–31 May 2027**.

---

1. Project Description

The problem

- **2.1 billion people** lacked safely managed drinking water in 2024, and 106 million still drank directly from untreated surface water (WHO/UNICEF JMP, 2025)
- Only about **2.5 %** of Earth's water is fresh, and of that, lakes hold just 0.26 % and rivers a fraction of a percent (USGS Water Science School)
- Agriculture takes about **70 %** of all fresh water withdrawn worldwide, and inefficient irrigation wastes much of it (FAO)
- About **126 billion m³** of piped water is lost or unaccounted for every year (Liemberger & Wyatt, *Quantifying the global non-revenue water problem*, *Water Supply*, 2019), and a leaking tap alone can waste thousands of litres a year (US EPA)

The causes are agriculture, the pollution of rivers and groundwater, and ever more frequent droughts. The numbers are global and abstract, though; nobody sees them in their own tap.

The solution

Five tools, one for each solution in the project brief:

1. **Irrigation lab** — pick a field size, a crop and how it is watered today, and see the water taken from the source with flood, sprinkler and drip irrigation, and what switching to drip saves (in Olympic swimming pools).
2. **Reuse loop** — follow the water from the taps to the treatment plant, where the pipe splits into what is reused and what is released. Set the city size, the water used per person and the share reused (with a shortcut for Israel's level).
3. **Solar desalination sizer** — how much electricity, how many solar panels (and football pitches of them) and how much brine a city's water needs, for three plant efficiencies and three sunny places.
4. **Leak hunter** — a game: a city loses 38 % of its water through six invisible leaks. Dig to find them, with and without pressure sensors, and count the dry holes.
5. **Home water audit** — shorter showers, tap aerators, a dripping tap and a rain barrel, each counted in litres. Add actions to a plan; the plan total follows you in the navigation.

Before the tools, the **Problem** page pours the planet into 100 litres (each dot is a litre) to show how little is drinkable, and lays out the three causes; the **Solution** page gives each fix with a real example and *the catch*.

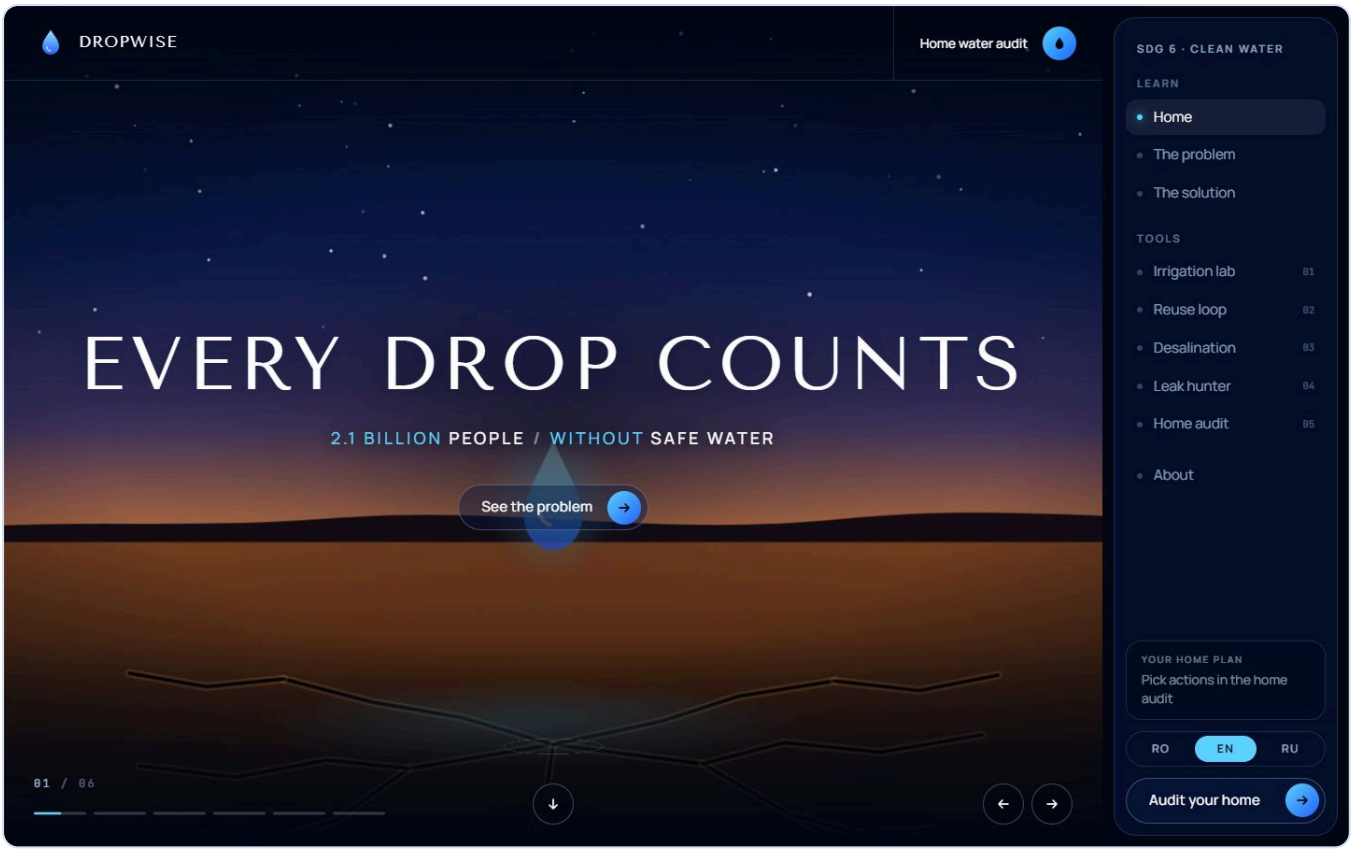
Available in **English, Romanian and Russian**. Everything you enter stays in your own browser; there are no accounts.

How it is different

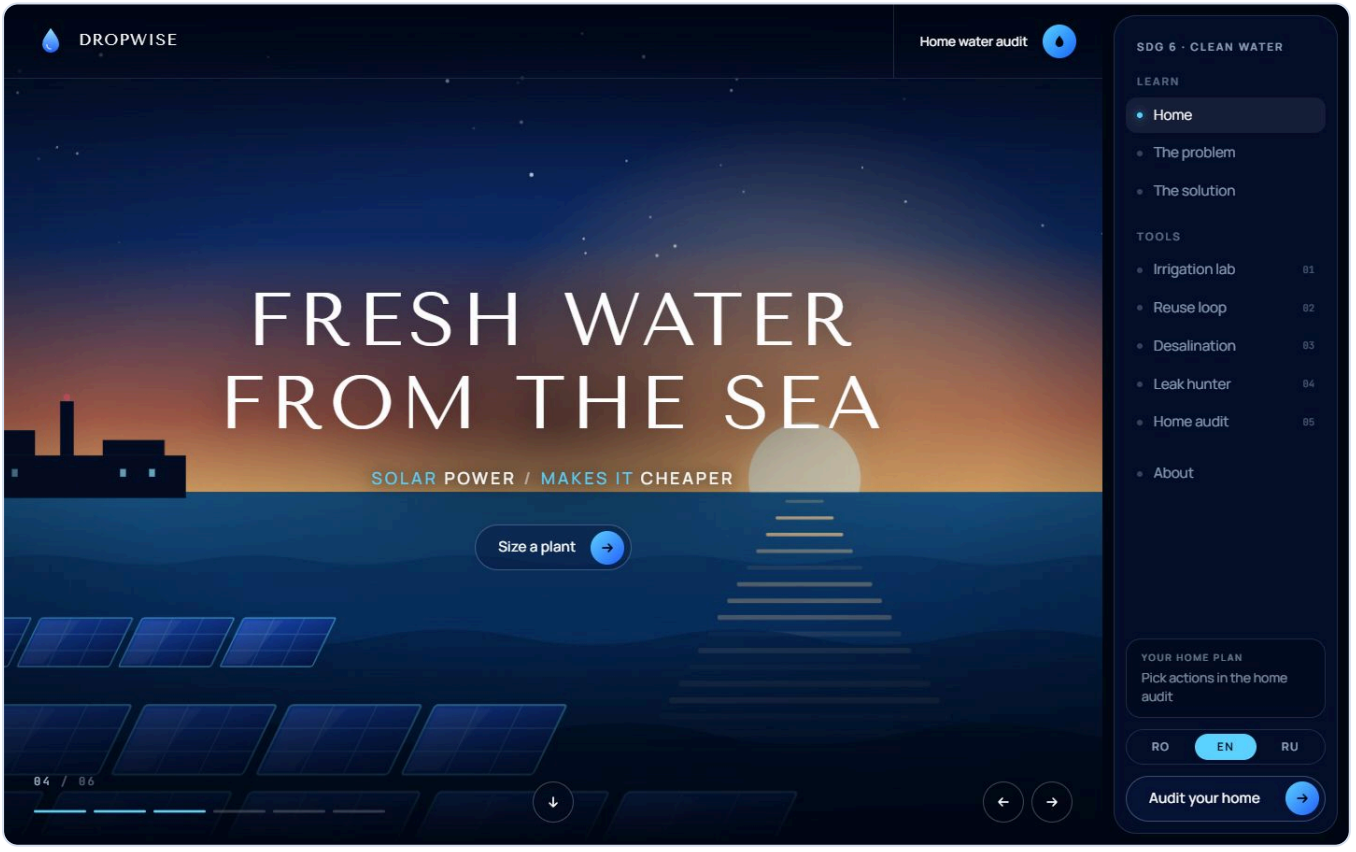
- **Honest numbers.** Every statistic names its source; every tool shows its formula; estimates are labelled estimates. Where sources differ (drip savings, Israel's reuse rate) the page says so.
- **Every solution comes with its catch** — cost, energy, brine, trust — instead of promising a miracle.
- **A slideshow home page** in the spirit of magnesia.cz: a full-screen hero with six animated scenes, a pinned horizontal row of solution cards, image breaks, numbered causes, counting statistics and an FAQ.
- **Navigation on the right**: a fixed panel on large screens, a drawer that slides in from the right on phones.
- **No tracking, no accounts, no backend to run.**

Screenshots

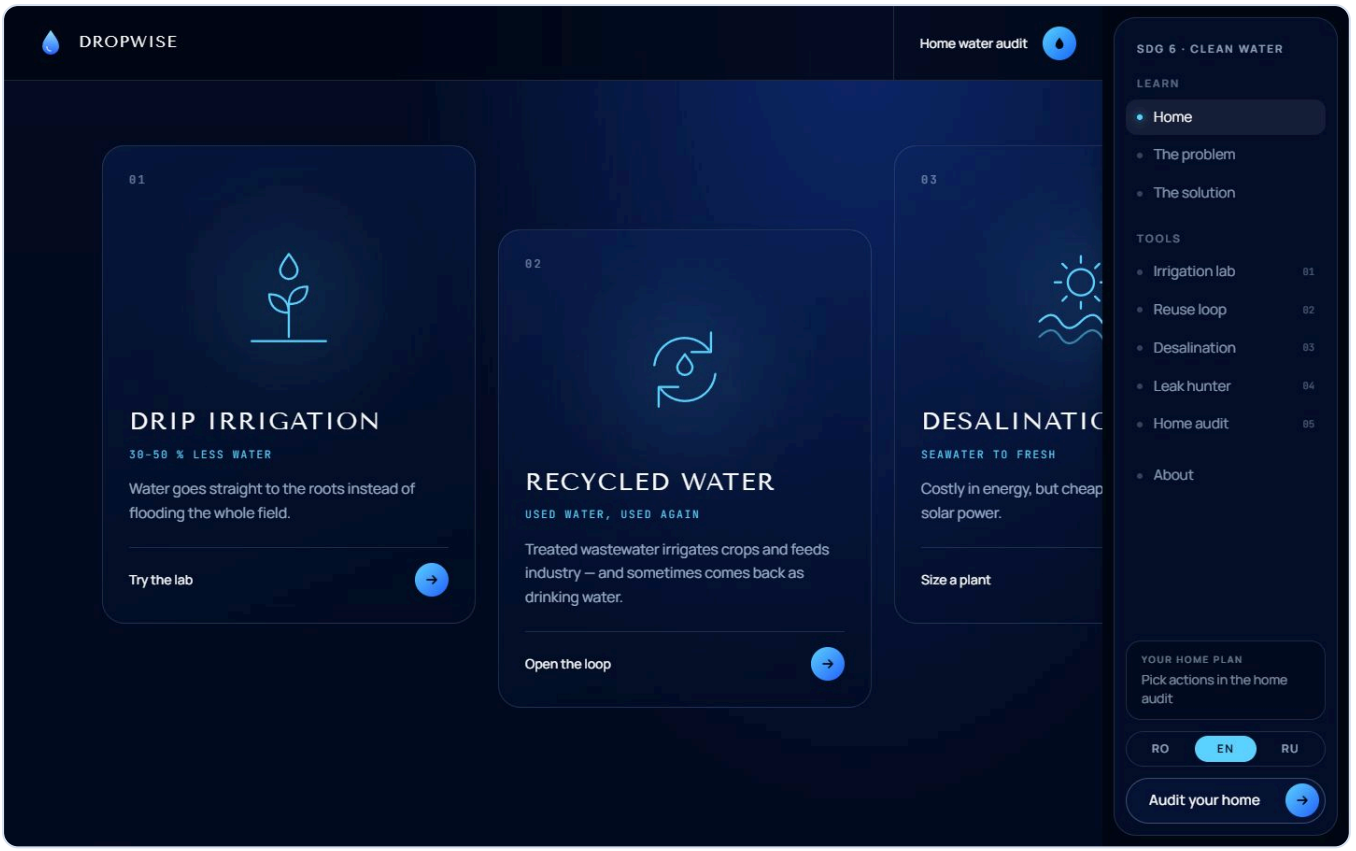
Taken from the running app at 1440 × 900 (phone shots at 390 × 844).



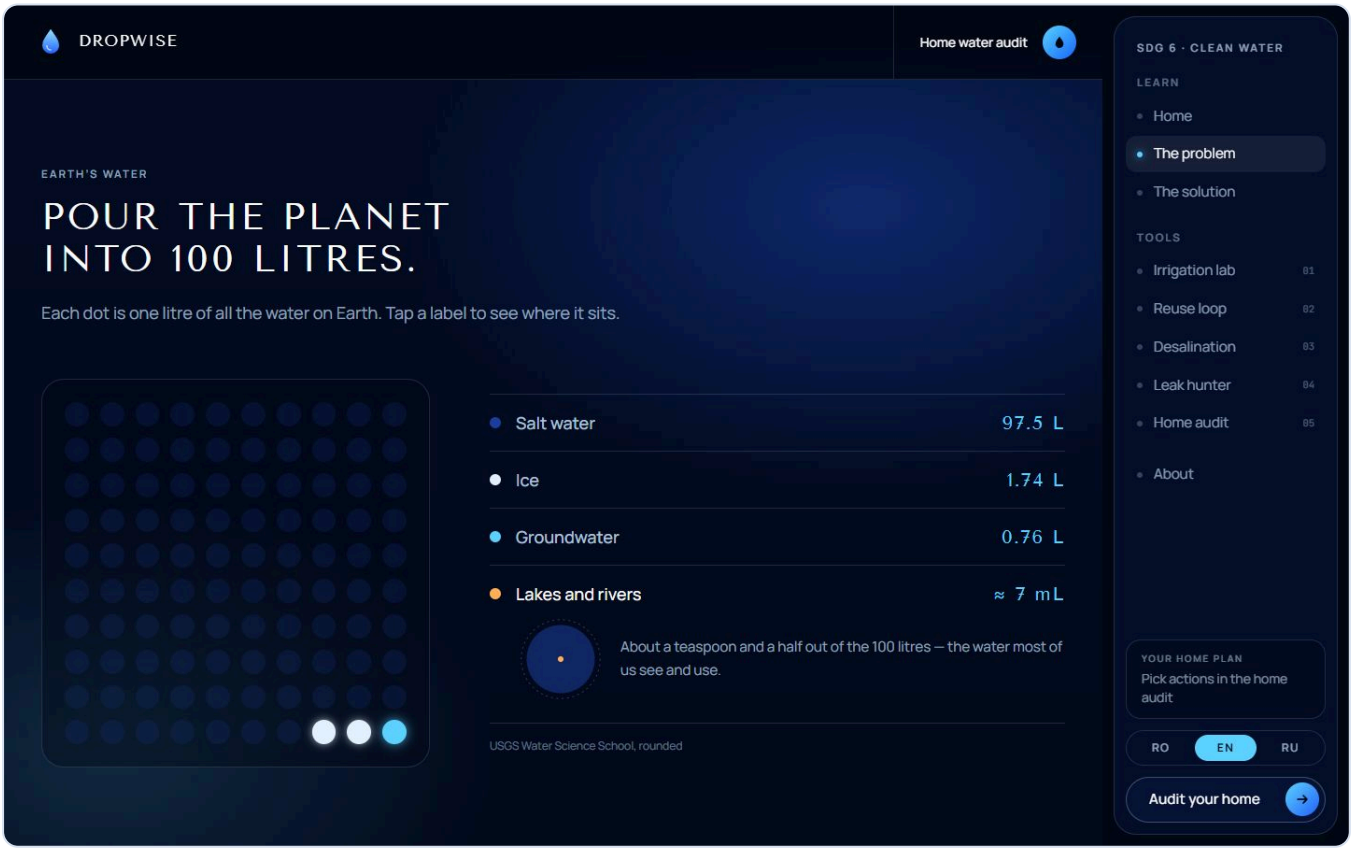
Home — the full-screen slideshow, the header with its quick action, and the navigation panel on the right.



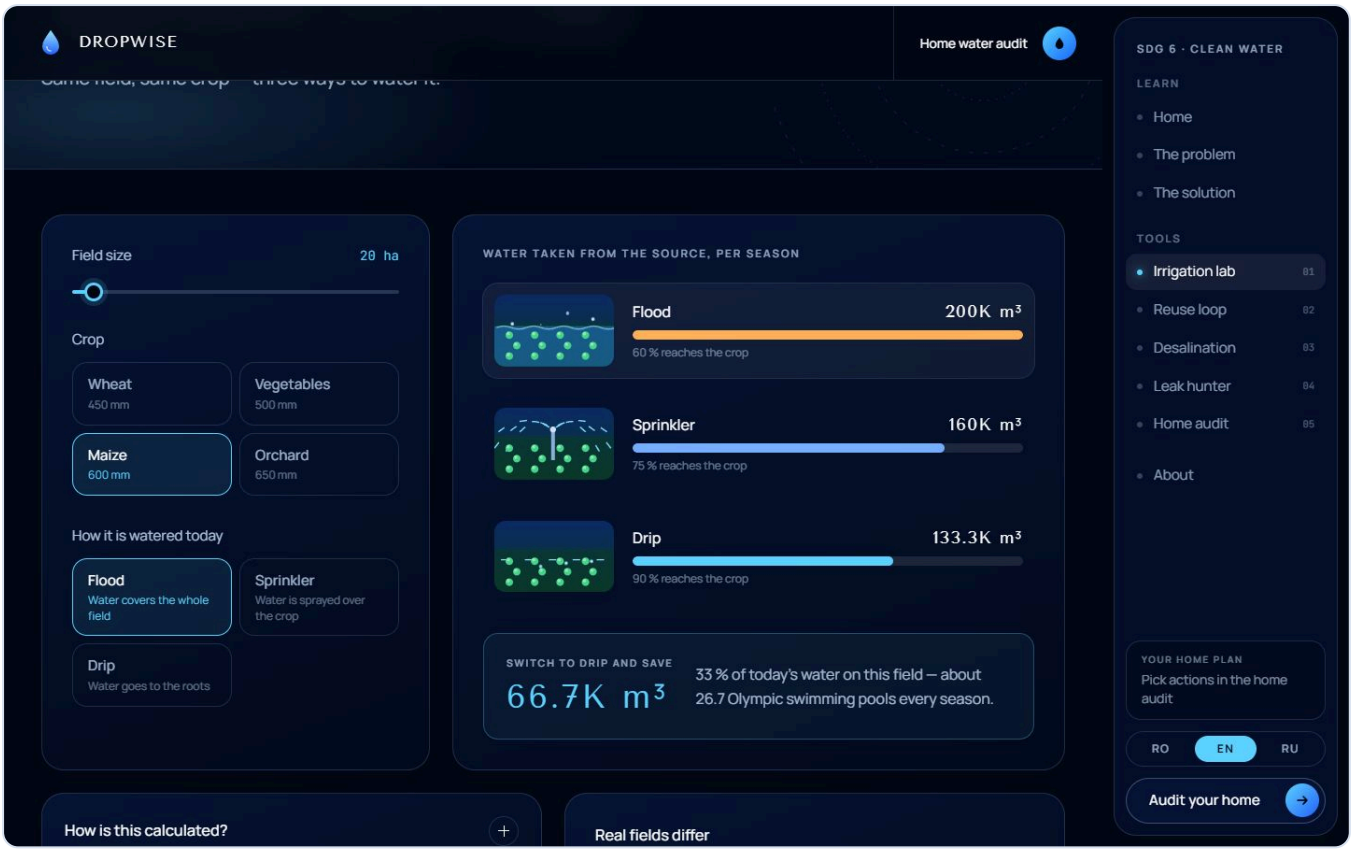
Slide 4 of 6 — every slide is an animated scene drawn in code, with a wide headline, a coloured tag line and a capsule button.



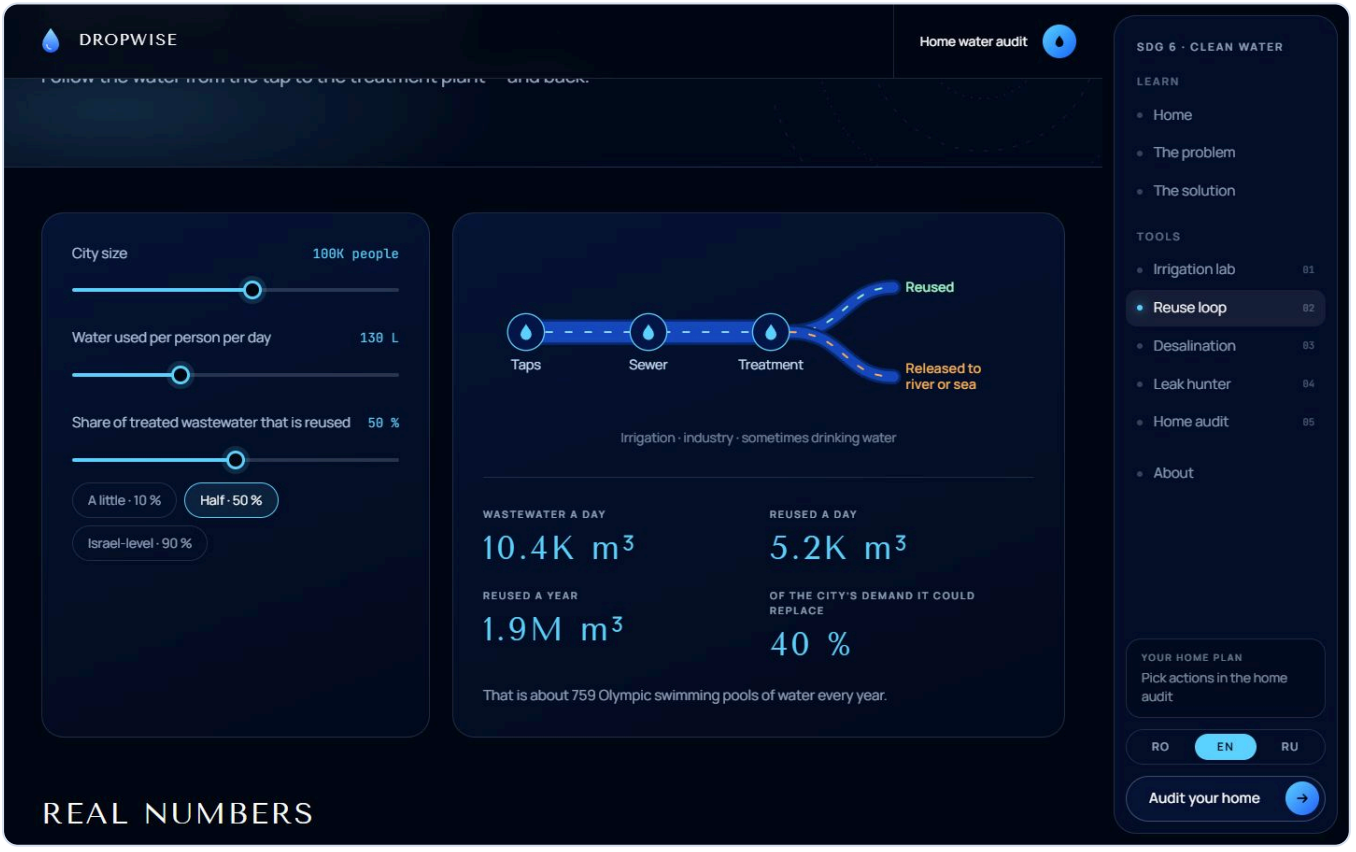
The five solutions on a track that slides sideways as the page scrolls, with the cards staggered like the product row on the reference site.



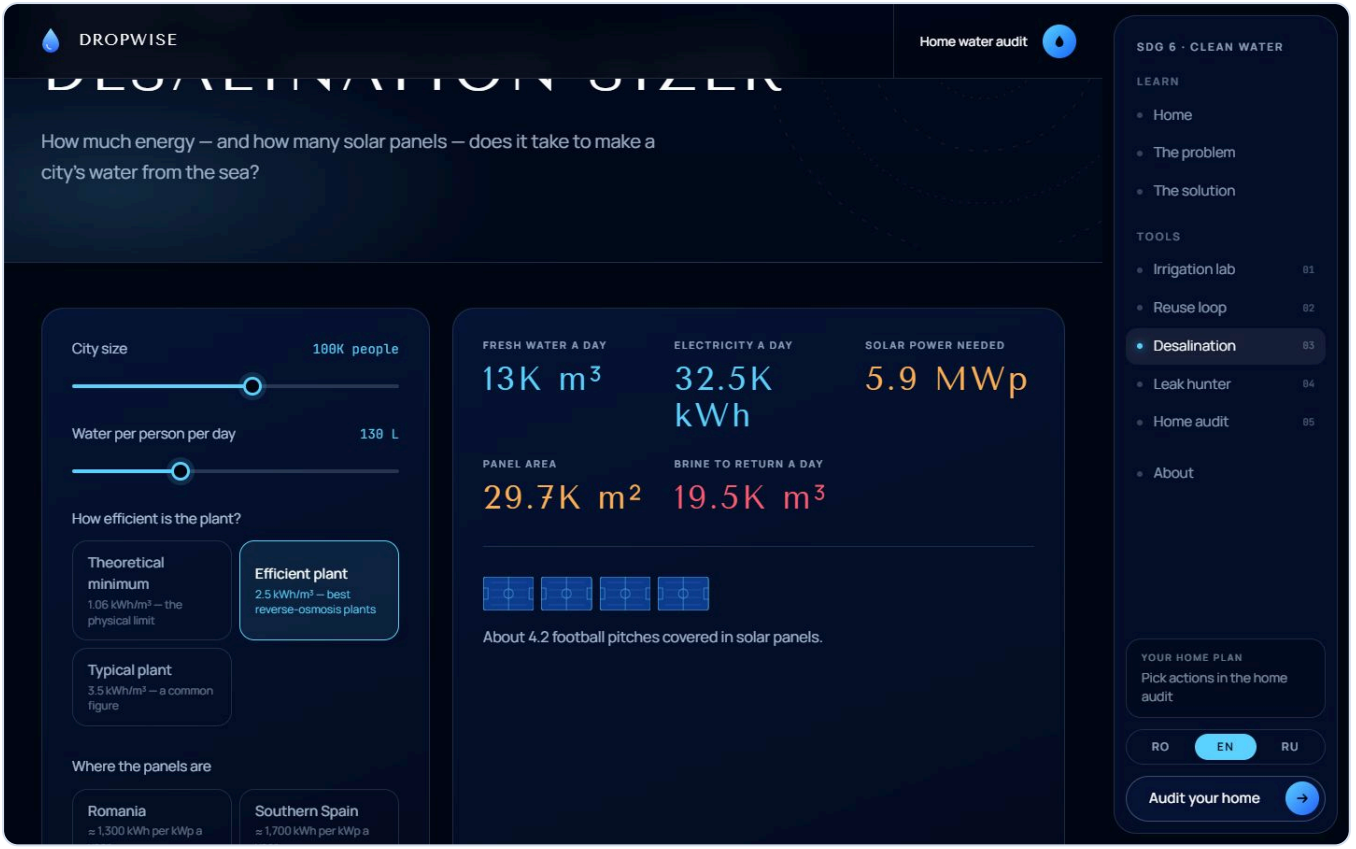
Problem — the planet poured into 100 litres. Lakes and rivers are drawn at their true size: about 7 mL, a speck inside one dot.



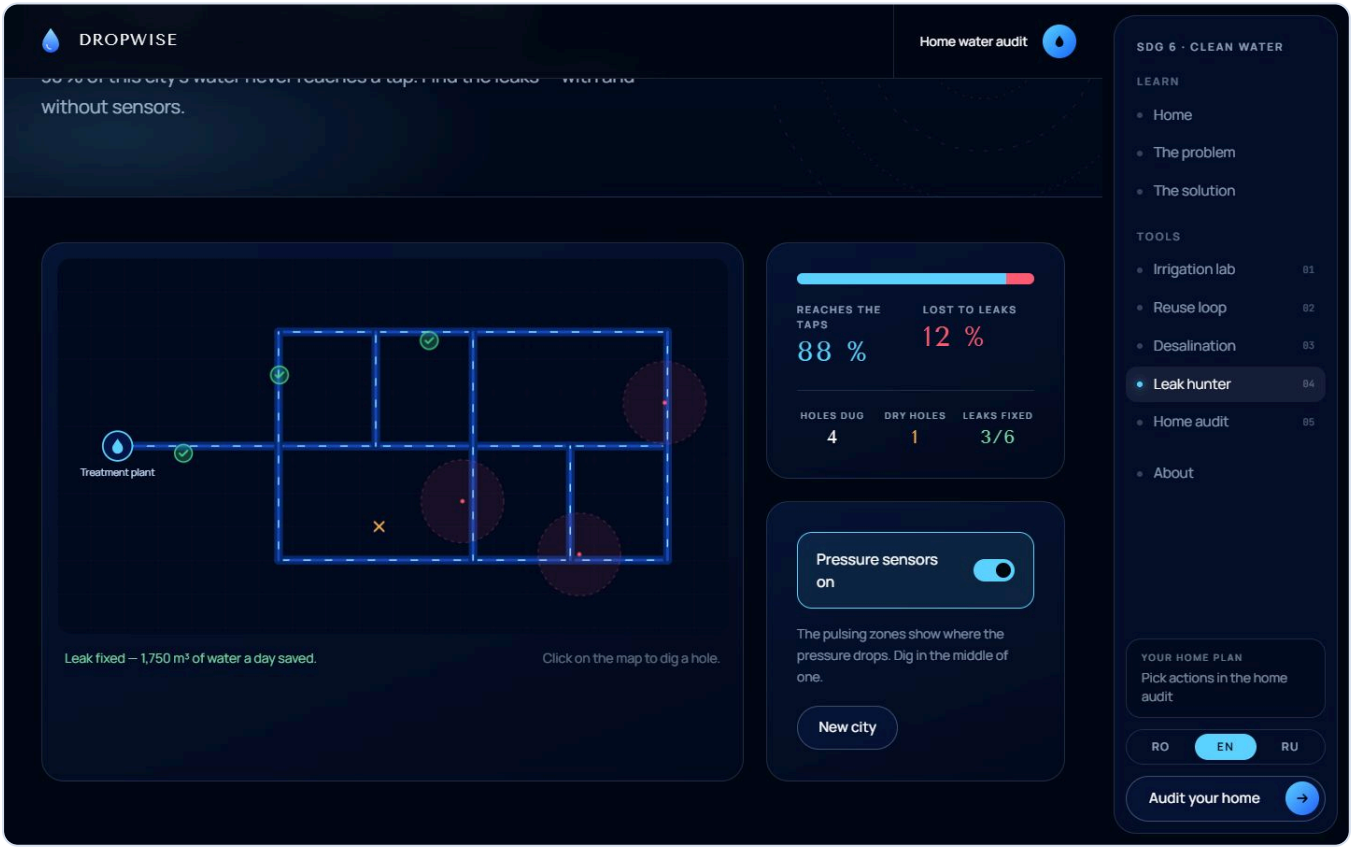
Irrigation lab — the same field watered by flood, sprinkler and drip, and what the switch to drip saves.



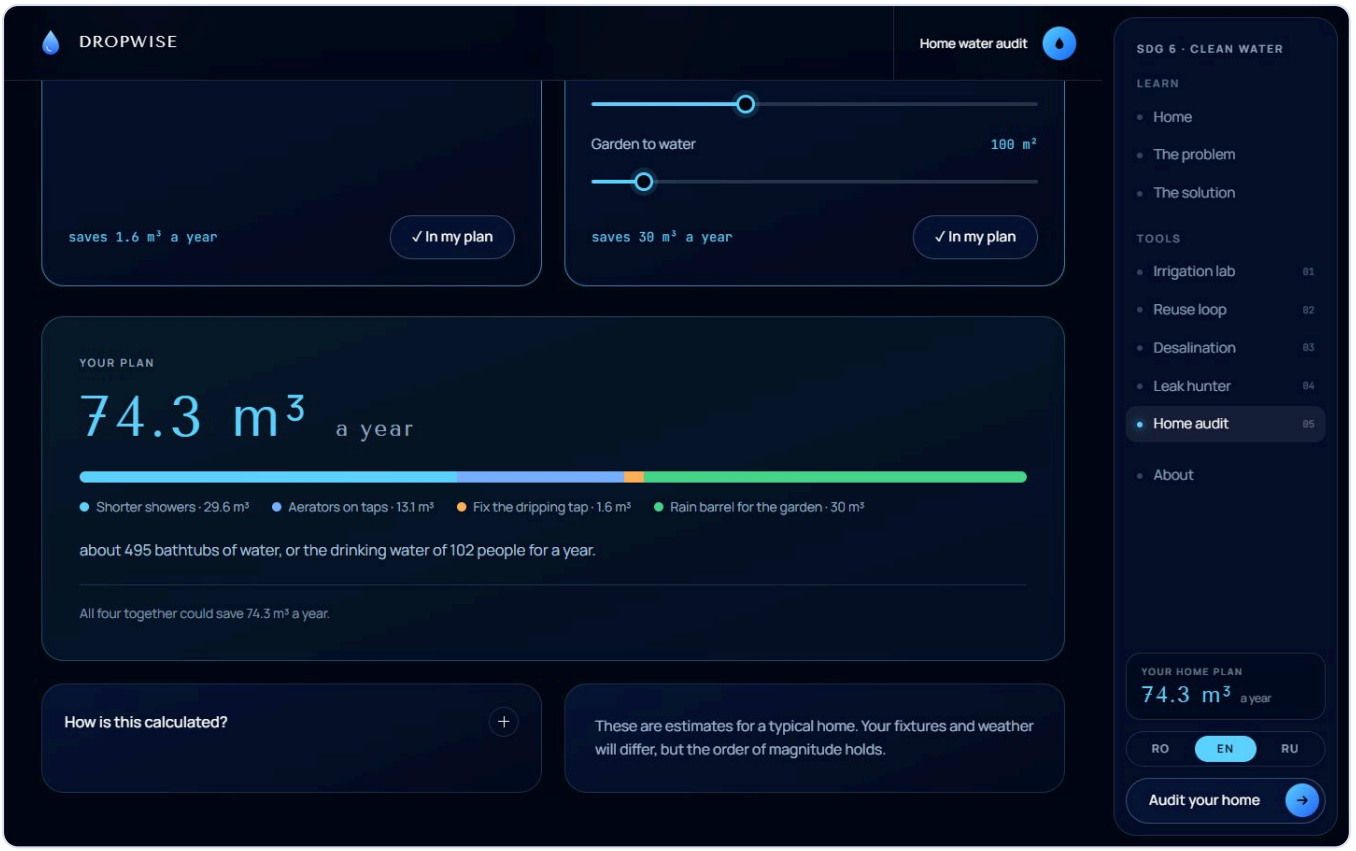
Reuse loop — the pipe splits into what is reused and what is released; the width follows the slider.



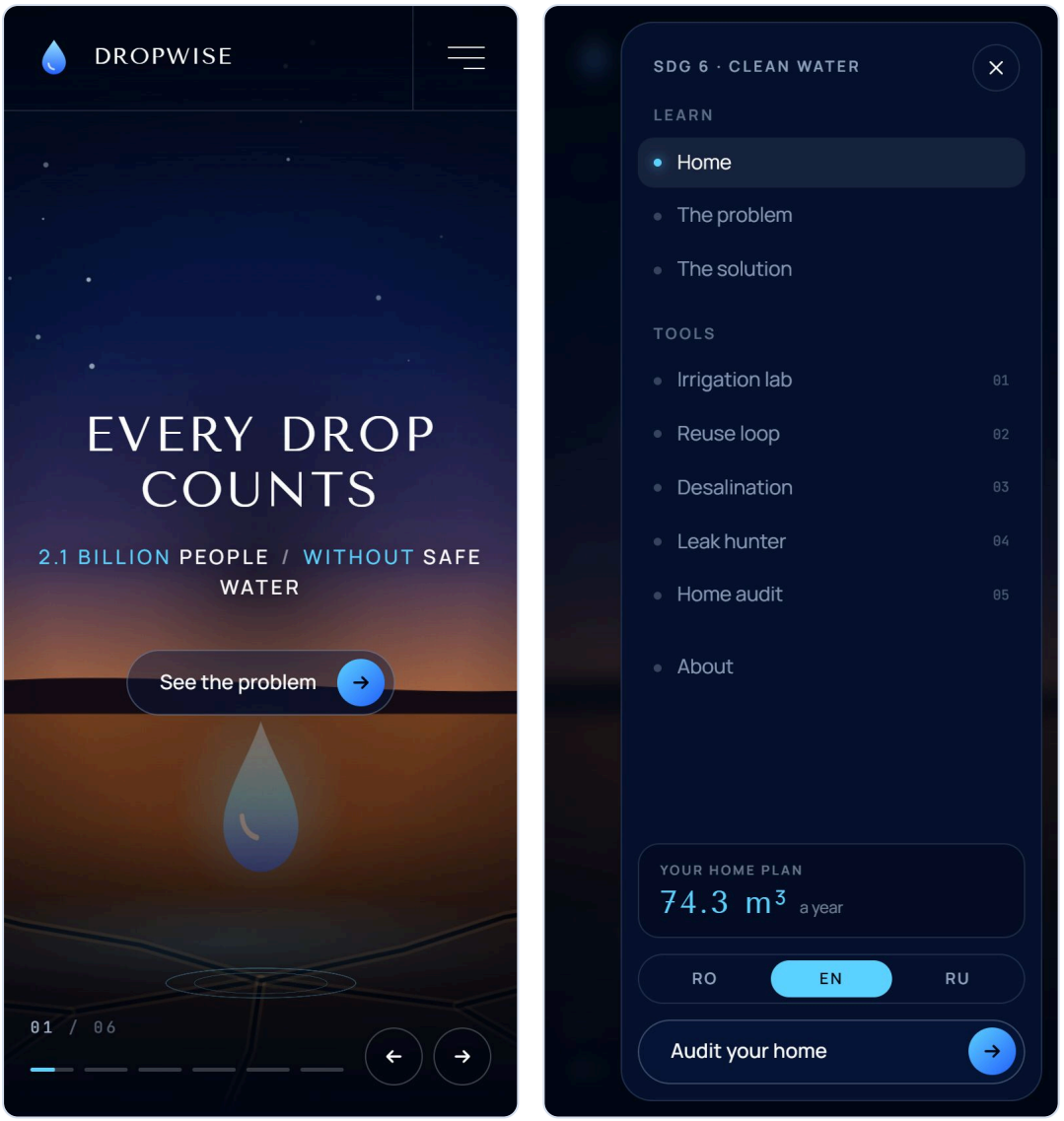
Solar desalination sizer — electricity, solar power, panel area in football pitches, and the brine to return.



Leak hunter — three of six leaks fixed with the help of pressure sensors, one dry hole, and the gauge falling from 38 % to 12 % lost.



Home water audit — four actions added to a plan; the total also shows in the navigation.



The same app on a phone — the menu button opens a drawer that slides in from the right.

---

2. People Involved

Role	Name
Student developer	Nicole Puzur (GitHub: nicolepuzur)
Supervisor	Azamat Jayush

Team size: 1 student (solo entry), within the category's 1–2 student limit.

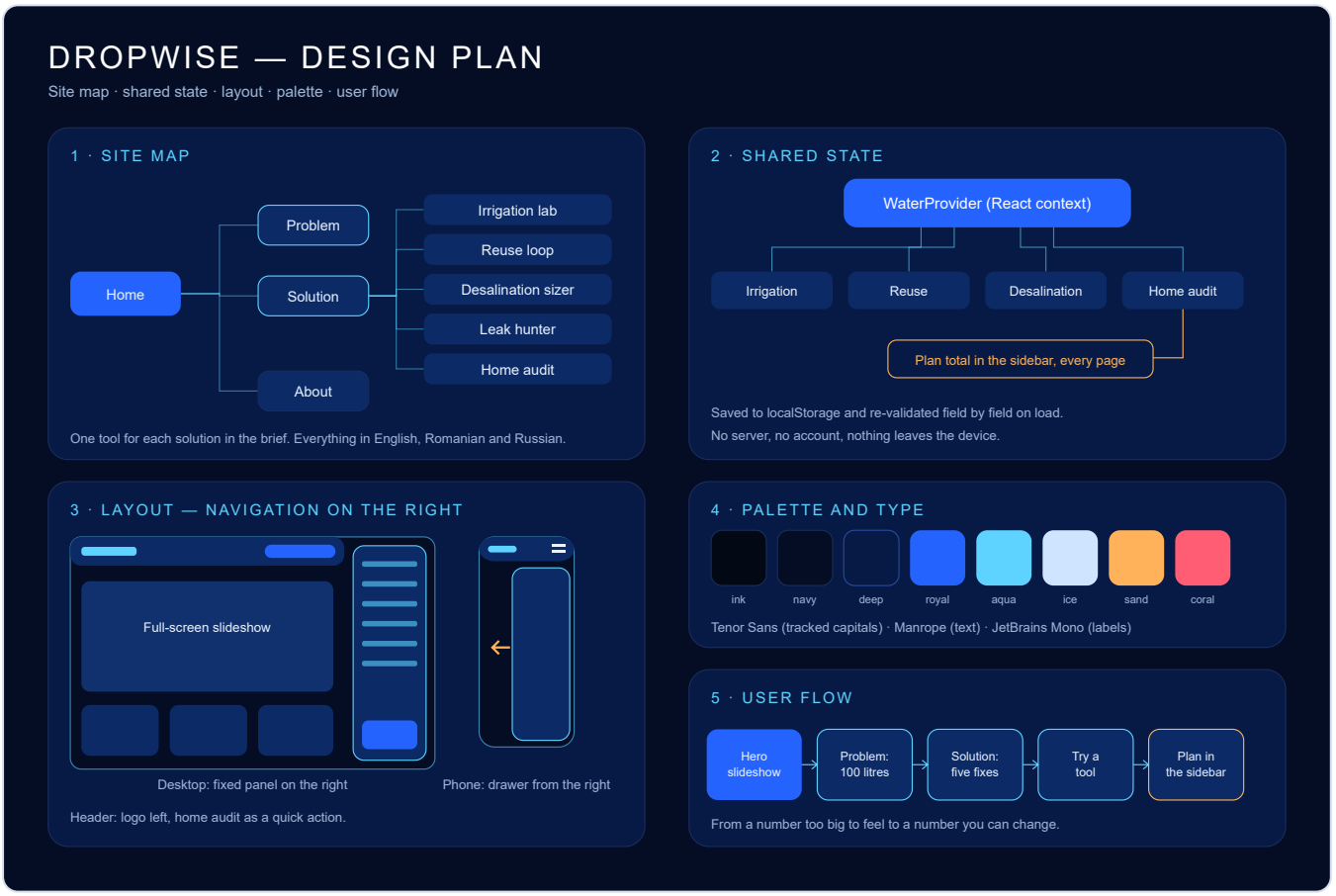
---

3. Early-Stage Plan, Design Decisions & Timeline

The project started from one framing question: *how do you make "2 billion people without safe water" something one person can feel and act on?* Three early decisions followed:

1. **One tool per solution.** The brief lists five solutions (agriculture, recycling, desalination, network losses, home), so each gets a small interactive model in which a person can change the numbers.
2. **Show the shortage before the fixes.** The 100-litre pour on the Problem page gives the scale in one picture, so the solutions have something to answer.
3. **Pure functions, tested.** The maths of every tool lives in a small tested module (`src/utils`), separate from the pages, so the numbers can be checked without the interface.

The structure follows the student's earlier projects (ClimaWatch, WasteLand and PlasticFree): a Problem page, a Solution page, interactive tools, an About page, English / Romanian / Russian and a test suite. The look follows a reference site, Magnesia: ink-navy sections lit by blue light, wide tracked capitals, a full-screen cinematic hero slideshow, capsule buttons with a luminous round arrow, a horizontal row of cards that slides sideways as the page scrolls, and smooth wheel scrolling. Only the *visual language* was taken; no code, images or fonts were copied. The reference's headline typeface (Aviano Sans) is commercial, so **Tenor Sans** is used instead, with **Manrope** for body text. Its photographs are replaced by scenes drawn in code. The navigation sits on the **right**.



Design plan — the site map, how the tools share one state, the layout with the navigation on the right (desktop and phone), the palette and the user flow.

Development timeline

Date	What happened
27 Oct 2026	**Research and plan.** Studied the structure of the earlier projects and the Magnesia reference; checked the key statistics against their sources (JMP, USGS, FAO, EPA, European Commission and others).

Date	What happened
27 Oct 2026	**Project and logic.** Vite + React + TypeScript scaffold, Tailwind v4 tokens, and the tested core: Earth's water, irrigation, reuse, desalination, leaks, home audit, number formatting and validated local storage.
27 Oct 2026	**Interface and content.** English, Romanian and Russian translations (with a test that they stay in step), the right-hand navigation, the six-scene hero slideshow and the pinned solution row, and every page.
27–28 Oct 2026	**Checks.** The main pages were looked at in a browser, then every page was run in English, Romanian and Russian at desktop and phone size with an automated layout check (no sideways scrolling, no text outside its box, no console errors); the fixes are listed in Problems faced.
27 Oct 2026	**Publication and documentation.** Public GitHub repository on the student's account, live deployment on Vercel, the team credited on the site, a design-plan sketch, this README as a PDF, and the AI usage log extended with a reproducible process. The README was checked against the official regulations, category page and rubric.

4. Architecture

```
Browser (React, no server of ours)
┌──────────────────────────────────────────────────────────────────┐
│  pages/       Home, Problem, Solution, Irrigation, Reuse,        │
│               Desalination, Leaks, AtHome, About                 │
│  components/  Header, Sidebar (right), HeroSlideshow, SolutionRail,  │
│               ImageBand, EarthWater, scenes/Scene (6 SVG scenes) ... │
│  context/     WaterProvider: the inputs of every tool, saved to  │
│               localStorage and validated on load                 │
│  utils/       earth · irrigation · reuse · desal · leaks · home ·    │
│               format · storage   (pure functions, unit-tested)   │
└──────────────────────────────────────────────────────────────────┘
        embeds only: Our World in Data chart, TED-Ed video (YouTube)
```

- **State.** One React context holds what the user enters. It is saved to `localStorage` and re-validated field by field on load (`utils/storage.ts`), so a corrupted or hand-edited entry can never crash the app.
- **One shared number.** The home audit's plan total (`utils/home.ts`) is shown in the navigation on every page.
- **Scenes.** The six hero scenes are SVG with CSS animations, drawn from seeded random numbers so they look the same on every visit. Slides that are out of view have their animations paused.
- **Pinned row.** On large screens the solution cards sit on a track inside a sticky container; the page's scroll position is turned into a sideways shift. On phones it is an ordinary swipeable row.
- **No backend.** There is nothing to hide and nothing to host.

5. Software & Hardware Specifications

Frontend	React 19, TypeScript, Vite, Tailwind CSS v4, React Router
Smooth scrolling	Lenis (MIT); off for reduced-motion users
Charts & animation	Hand-written SVG and CSS — no chart or animation library

Chart embed	Our World in Data
Translations	Custom React context, English / Romanian / Russian, no external library
Typography	Tenor Sans, Manrope and JetBrains Mono (Google Fonts, SIL Open Font License)
Testing	Vitest
Hardware	None. DropWise is a web application; it runs in any modern browser on any device with an internet connection.

Project structure

```
src/
  components/      Header, Sidebar, SidebarBody, HeroSlideshow, SolutionRail, ImageBand,
EarthWater, ...
    scenes/        Scene.tsx — the six animated SVG scenes
  pages/           Home, Problem, Solution, Irrigation, Reuse, Desal, Leaks, AtHome, About
  context/         WaterProvider (shared state)
  hooks/           useCountUp, useInView, useMediaQuery, useScrolled
  i18n/            en / ro / ru translations + language context
  utils/           earth, irrigation, reuse, desal, leaks, home, format, storage, scroll (+ tests)
  constants/       Routes, external links
```

---

6. Problems Faced & How They Were Solved

- **Sources disagree on some numbers.** Drip savings are quoted as 30–50 % in some places and up to 40 % in others; Israel's reuse is reported between about 85 and 90 % depending on how wastewater is counted. *Solution:* the pages give the range, name the sources and say the figure depends on the method; the irrigation lab computes its own result from stated efficiencies instead of repeating a headline.
- **Not every figure could be confirmed at the primary source.** Singapore's "up to 40 %" for NEWater was found in secondary reporting, and the PUB page did not repeat it. *Solution:* it is worded as "reported", attributed to PUB, and flagged in the README sources so it can be re-checked before submission.
- **Unit symbols in capital letters.** The headline style turns text to uppercase, which made cubic metres read "M³" and kilowatt-hours "KWH". *Solution:* numeric displays opt out of the uppercase style.
- **A wrong unit for megawatts.** The first version glued an "M" in front of "kWp" and wrote "MkWp". *Solution:* a proper `MWp` unit in every language.
- **Dots spilling out of their frame.** The 100-litre grid used percentage gaps, which collapse in an auto-height container and let the dots spill out of their frame. *Solution:* fixed rem gaps.
- **Grey blocks before a section revealed itself.** Cells that fade in on scroll showed the divider colour behind them. *Solution:* the cells carry their own borders over a plain background.
- **Text on a bright sunset.** The tag line under the "Fresh water from the sea" headline disappeared against the glow. *Solution:* a soft dark vignette behind the text and a text shadow.
- **A drop falling through the headline.** The tap scene's drop crossed the centred text. *Solution:* the faucet moved to the right third, and that scene is anchored to the right edge on narrow screens.
- **Long Russian and Romanian words in wide capitals.** The automated layout check found that the heading of the sliding solution row and the card titles were wider than their boxes in Romanian and

Russian. *Solution:* smaller, fluid sizes for those headings. The texts themselves are still to be proof-read by a fluent reader before submitting.

- **A form wider than a phone screen.** On the Russian irrigation page the whole page scrolled sideways by 30 px, because a `<fieldset>` is as wide as its longest content by default. *Solution:* `min-w-0` on fieldsets and single-column grids that may shrink.
- **The giant footer name was cut off.** It was sized from the window width, but on large screens the page is narrower by the width of the right-hand panel, so the last letters fell outside. *Solution:* size it from the space it really has.
- **A pinned section that must not trap people.** *Solution:* the sideways row is only pinned on large screens and for people who have not asked for reduced motion; everyone else gets a normal swipeable row.
- **Corrupted or hand-edited saved data.** *Solution:* `sanitizeState` rebuilds the state from whatever it finds and falls back to defaults field by field (tested).
- **A right-hand navigation on a phone.** *Solution:* a fixed panel from the `lg` breakpoint up, and below it a header with a menu button that opens a drawer from the right (closes on route change and on Escape, locks scrolling while open).
- **A commercial typeface.** The reference uses Aviano Sans. *Solution:* Tenor Sans, a free wide humanist sans, set in tracked capitals.

---

7. Responsible AI Usage

AI tools were used during development, in line with the category's rules on citing AI assistance.

- **Tool used:** Claude Code (Anthropic), an AI coding assistant, used as a pair-programming tool.
- **Model:** Claude Sonnet 5.5 (`claude-sonnet-5-5`).
- **Prompts:** every prompt is quoted, with what it produced and how it was checked, in docs/AI_USAGE.md.
- **How it was used:** scaffolding the project on the structure of the student's earlier projects, writing the TypeScript, drawing the scenes in code, generating the tests, translating the interface and drafting this README, from the subject text and design direction given by the student.
- **Human oversight:** generated code is run, type-checked, linted and tested before being kept, and the pages were checked in a browser. Product decisions (the topic, the five solutions, the right-hand navigation, the reference design) were directed by the student.
- **Reproducible process:** docs/AI_USAGE.md lists the steps, tools and commands needed to rebuild the project from the same brief.
- **Licensing and privacy:** the code is MIT-licensed, third-party content and its licences are listed below, fonts are open-licence, and the site stores only the visitor's own choices in their browser (no cookies, no analytics, no accounts).
- **What was not AI-generated:** the statistics and their sources, the Our World in Data chart and the TED-Ed video are real external sources, credited below. There are no AI-generated images; the scenes are drawn in code.

No AI-generated text, image or code in this project is presented as original human work without this disclosure.

Running locally

```
npm install
npm run dev
```

No API key or `.env` file is needed.

Testing

```
npm test
```

52 tests cover Earth's water (the shares add up to 100, dots are handed out exactly, 7 mL of lakes and rivers), the irrigation lab (efficiencies, the third saved by drip), the reuse loop (wastewater, split between reused and released), the desalination sizer (energy, panels, brine, bad input), the leak hunter (every leak is on a pipe, digging, sensor hints), the home audit (every action and its caps), number formatting, the validation of saved data, and a check that the English, Romanian and Russian texts have the same keys, list lengths and placeholders.

```
npm run lint
npm run build
```

Deployment

A static single-page application. The live app runs on Vercel, connected to this repository: every push to `main` deploys automatically. To host your own copy, import the repository in Vercel (or run `npx vercel --prod`) and deploy; `vercel.json` rewrites every route to `index.html` so direct links to inner pages work. No environment variables are needed.

Sources

WHO/UNICEF Joint Monitoring Programme, *Progress on household drinking water, sanitation and hygiene 2000–2024* (2025, washdata.org); USGS Water Science School, *Where is Earth's water?* (usgs.gov); FAO AQUASTAT and irrigation guidance (fao.org); Our World in Data, *Agricultural water as a share of total water withdrawals* (ourworldindata.org); Liemberger & Wyatt, *Quantifying the global non-revenue water problem*, Water Supply (2019); US EPA WaterSense, *Fix a Leak Week* (epa.gov); Elimelech & Phillip, *The future of seawater desalination*, Science (2011); United Nations University, global brine production from desalination (2019); European Commission, water reuse (environment.ec.europa.eu); Israel Water Authority and PUB Singapore (reported figures); Project Drawdown, farm irrigation efficiency (drawdown.org); United Nations Sustainable Development Goal 6 (sdgs.un.org/goals/goal6); TED-Ed, Christiana Z. Peppard, *Fresh water scarcity: an introduction to the problem*.

Third-party content & licences

Content	Licence / terms	How it is used
Tenor Sans, Manrope, JetBrains Mono	SIL Open Font License	Loaded from Google Fonts
Lenis	MIT	Smooth scrolling
Our World in Data chart	CC BY 4.0	Embedded, with credit
TED-Ed lesson by Christiana Z. Peppard	YouTube embed	Embedded (privacy-enhanced `youtube-nocookie.com`), with credit
Statistics: WHO/UNICEF, USGS, FAO, Liemberger & Wyatt, EPA, European Commission	Cited facts	Quoted with their source next to each figure
React, Vite, Tailwind CSS, React Router	MIT	Dependencies (see `package.json`)

No photographs or illustrations from other sources are used: the scenes are drawn in code.

License

MIT, © 2026 Nicole Puzur. Open source, per Infomatrix competition rules.

Appendix A — AI usage log

Infomatrix asks Software Development entrants to *"cite all prompts, models, or AI resources used and give proper credit in their documentation and presentation."* This file is that citation: what was used, the prompts, what the AI produced, and how a human checked it.

Tool and model

Tool	Claude Code, Anthropic's AI coding assistant (desktop app)
Model	Claude Sonnet 5.5 (`claude-sonnet-5-5`), Anthropic
Role	Pair programmer: wrote code, tests, translations and documentation from the student's instructions
Not used	No AI-generated images. The scenes are drawn in code. Statistics come from WHO/UNICEF, USGS, FAO, the US EPA and others; the chart is from Our World in Data and the video is a TED-Ed lesson.

Prompts

The student's prompts, as typed (spelling kept):

1. ONE MORE SITE WITH ANOTHER PROBLEM SAME LIKE THE PREVIOUS ONES BUT NAVBAR ON THE RIGHT THIS IS THE SUBJECT:

   Problema: lipsa apei potabile Doar o parte foarte mică din apa planetei e dulce și accesibilă. Peste 2 miliarde de oameni nu au acces sigur la apă potabilă. Cauzele principale sunt agricultura (cel mai mare consumator, cu irigații ineficiente), poluarea râurilor și a pânzei freatice, și secetele tot mai dese din cauza schimbărilor climatice. Soluții

   1. Agricultură: irigarea prin picurare duce apa direct la rădăcină și poate reduce consumul cu 30–50% față de inundarea câmpului. Israelul a ajuns lider mondial tocmai așa.
   2. Reciclarea apei uzate: apa din canalizare e tratată și refolosită pentru irigații sau industrie, iar uneori chiar ca apă potabilă. Singapore (sistemul NEWater) și Israel (aproape 90% din apa uzată refolosită) sunt exemple bune.
   3. Desalinizare: transformă apa de mare în apă dulce. E scumpă energetic, dar devine mai ieftină dacă e alimentată cu energie solară. Arabia Saudită și Emiratele depind masiv de ea.
   4. Pierderi din rețea: în multe orașe, inclusiv din România, o mare parte din apă se pierde prin țevi vechi, uneori 30–40% sau chiar mai mult. Senzorii de presiune și înlocuirea conductelor rezolvă mult fără să ai nevoie de apă „nouă”.
   5. Acasă: dușuri scurte, robinete cu aerator, colectarea apei de ploaie pentru grădină și repararea scurgerilor. Un robinet care picură poate pierde mii de litri pe an.

   SITE SHOULD HAVE THIS TYPE OF DESIGN: https://www.magnesia.cz LIKE THIS TYPE OF SLIDESHOW. AND THE PROJECT SKELEYPN SHOULD LOOK LIKE THE PREVIOUS ONES (CLIMA WATCH, WASTELAND, AND THE ONE WITH PLASTIC POLLUTION)

2. ok but i will do it on a different Git account and vercel account too

3. OK LETS CONNECT IT OT MY GITHUB ACCOUNT

4. im on nicoles account

5. keep the repo public the teach has to see it

6. check and do the guidelines for this site: https://www.infomatrix.ro

   name: NICOLE PUZUR

   TEACHER : AZAMT JAYUSH

Prompts that only dealt with which GitHub account to use are left out (account administration, not project content). The supervisor's name is written "Azamat Jayush" in the documentation, as in the student's earlier projects.

What the AI did with them

- Read the student's earlier projects (ClimaWatch and PlasticFree) to reuse their structure: Home, Problem, Solution, interactive tools, About, three languages, a shared state saved in the browser, tested utility modules, a Vercel rewrite file and an Infomatrix-style README.
- Looked at the Magnesia reference site and took its visual language: a full-screen hero slideshow, ink-navy sections with blue light, wide tracked capitals, capsule buttons with a round arrow, a pinned horizontal row of cards, image breaks, numbered causes, an FAQ and smooth scrolling. No code, images or fonts were copied.
- Put the navigation on the right, as asked.
- Turned the five solutions in the brief into five tools: irrigation lab, reuse loop, solar desalination sizer, leak hunter, home water audit.
- Wrote the code, the six animated scenes (SVG), the tests, the English, Romanian and Russian text and the documentation.
- Kept the project in its own folder, apart from the student's other projects, as the second prompt asked.
- Published the code to the student's own GitHub account (`nicolepuzur`), after she signed in herself in the terminal. The Vercel deployment was done by the student in her own Vercel account.
- Read the official Infomatrix regulations, the Software Development page and the evaluation rubric, and brought the documentation in line with them: the checklist and the table for the nine judging criteria in the README, the team credit on the About page, a design-plan sketch, a PDF copy of the documentation, this log, and `docs/SUBMISSION.md` .

What was checked, and what is left for the student

These checks were run by the AI assistant, with web look-ups, tests and a browser, while building. They are not a substitute for the student's own review (see the end of this section).

- **Statistics were looked up at their sources** before being used: 2.1 billion without safely managed drinking water in 2024 and 106 million drinking untreated surface water (WHO/UNICEF JMP 2025); 2.5 % of Earth's water is fresh and lakes hold 0.007 % of all water (USGS); about 70 % of withdrawals go to agriculture, up to 95 % in some countries (FAO); 126 billion m^3 of non-revenue water a year (Liemberger & Wyatt); a dripping tap wasting more than 3,000 gallons a year and a household leak more than 9,300 gallons (US EPA); about 1 billion m^3 of urban wastewater reused in the EU, with six times more possible (European Commission); the 1.06 kWh/m^3 minimum and 2.5–4 kWh/m^3 for real plants (Elimelech & Phillip, 2011).
- **Figures that came from the student's brief** and could not all be confirmed at a primary source: "30–50 %" for drip irrigation (field studies and Project Drawdown give about 30–40 %, so the site says the range varies), Israel reusing "almost 90 %" (reports give 85–90 %), and old networks in Romania losing 30–40 %. They are worded as ranges or as reported figures.
- **Singapore's NEWater share (up to 40 %)** was found in secondary reporting only; the PUB page checked did not repeat it. The site words it as "reported" and the README flags it for a last check before submission.
- **Assumptions are labelled**: drip, sprinkler and flood efficiencies (60 / 75 / 90 %), the crop water needs, the share of used water that returns to the sewer (80 %), shower and tap flows, the size of a drop, the solar yield of three regions and the size of a panel. Each tool shows its formula in a "How is this calculated?" box.
- Every calculation is a small function with unit tests (52 tests). The pages were run in a browser: the main pages were looked at by eye, and then all nine pages plus the "not found" page were run in English, Romanian and Russian, at desktop and phone size, with an automated layout check (no sideways scrolling, no text outside its box, no console errors). That check found four layout problems, which were fixed (see the README, *Problems faced*). The check script is a one-off and is not part of the repository.
- The English, Romanian and Russian texts are checked by a test for identical keys, list lengths and placeholders. A fluent reader should still read the Romanian and Russian once.

Still for the student to do

- Read every page in English, Romanian and Russian, and change anything that does not sound right. The layout was checked automatically, but no fluent reader has proof-read the Romanian or Russian text.
- Re-open the sources behind the key statistics, and the Singapore figure, before submitting.
- Run `npm run dev`, try each tool, and be able to explain its formula (each one is in its "How is this calculated?" box and in the README).
- Add the live address to the README, and confirm the documentation with the supervisor before uploading it.

Reproducible process

Anyone can rebuild the project from the same brief. The result will be similar but not identical, because AI output varies between runs; the tests, the formulas and the cited sources are what make the result checkable.

1. **Tools:** Claude Code (desktop app) with the model `claude-sonnet-5-5`; Node.js 24; a browser for checking.
2. **The brief:** prompt 1 above is the whole brief: the subject text, the reference site and "same structure as the previous projects, navigation on the right".
3. **Research first:** the AI read the earlier projects' structure (shell, navigation, i18n, calculation modules, tests, README) and the reference site's sections and slideshow, and checked the statistics against their sources with web searches.
4. **Build in layers**, checking after each one:
    1. configuration (`package.json`, TypeScript, Vite, Tailwind, Vercel rewrite);
    2. pure calculation modules in `src/utils` with their tests;
    3. English, Romanian and Russian text, with a test that keeps the three in step;
    4. shared components (navigation, slideshow, scenes, controls), then the pages;
    5. documentation, screenshots and this log.
5. **Commands used after each layer:** `npx tsc -b` (types), `npm run lint`, `npm test`, `npm run build`, and `npm run dev` to look at the pages in a browser.
6. **Screenshots:** taken from the running app with a headless browser at 1440 × 900 and 390 × 844.
7. **Human decisions:** the topic, the five solutions, the reference design, the right-hand navigation, publishing under the student's own account and keeping the repository public were the student's.

Privacy and licensing

- **Privacy:** the site stores only the visitor's own choices (language, tool inputs, home plan) in their browser, with no cookies, no analytics and no accounts. Loading the page contacts Google Fonts, and the Problem page loads a chart from Our World in Data and a video from YouTube's privacy-enhanced domain.
- **Licensing:** the code is MIT-licensed; the fonts are under the SIL Open Font License; Lenis is MIT; the Our World in Data chart is CC BY 4.0; the TED-Ed video is embedded, not copied. No images from other sources are used, and the reference site's code, images and fonts were not copied. The full table is in the README.